# Quan Tran Hong

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## Research Interests

Machine Learning, Computer Vision, 3D Reconstruction, Neural Rendering, Generative Modeling.

## Education

**VinUniversity** Mar 2026 – Mar 2028
*Master of Science in Computer Science* (Expected)
*Graduate Research Excellence Program: full-tuition scholarship and monthly stipend.*

**Auckland University of Technology** Sep 2019 – Aug 2023
*Bachelor of Computer and Information Sciences*

## Research Experience

**Research Assistant**, *Center for AI Research, VinUniversity* Nov 2025 – Present
Supervisor: Asst. Prof. Binh T. Nguyen

- Developed a variational approach to learning compact 3D representations, with research focused on optimization and probabilistic machine learning for 3D vision.
- First author accepted at NeurIPS 2026 for poster presentation.

**Research Assistant**, *Cognitive Robotics Lab, University of Arkansas* Dec 2024 – Nov 2025
Supervisor: Asst. Prof. Tuan Dang

- Proposed and developed geometric regularization for 3D reconstruction; first author of TriaGS.
- Co-first-authored research on geometry-aware mapping for visual SLAM.

## Publications

[1] **Q. T. Hong**, L. Nguyen-Chi, and B. Nguyen. "Birth-Death Structural Learning for 3D Gaussian Splatting." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. *Accepted as a poster*. [Paper] [Code]

[2] C. T. Luu[†], **Q. T. Hong**[†], L. H. Phan, and T. Dang. "Geometry-Aware Online Mapping for 3D Gaussian Splatting SLAM." *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. *Accepted*. [Paper] [Code]

[3] **Q. T. Hong** and T. Dang. "TriaGS: Differentiable Triangulation-Guided Geometric Consistency for 3D Gaussian Splatting." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. [Paper] [Code]

[4] **Q. T. Hong**, L. Nguyen-Chi, and B. Nguyen. "Applying Splat Regression Models to Particle Density Control in Radiance Fields." *Accepted to the ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM)*. [Paper]

[5] **Q. T. Hong** and B. Nguyen. "Render-Aware Subset Selection for Post-Training 3D Gaussian Splatting Compaction." Manuscript. [Paper]

[6] H. K. L. Tran[†], **Q. T. Hong**[†], L. Dinh, M. T. Thai, and T. B. Nguyen. "Safety Steering of Vision-Language-Action Models with Success-Guided Barrier Projection." *Submitted to the IEEE International Conference on Robotics and Automation (ICRA)*, 2027.

[7] Q. Nguyen, N. B. T. Bac, **Q. T. Hong**, A. T. Luu, V. K. Vu, and B. Nguyen. "Mitigating Hallucination in Visual Language Models with Partial Transport KL-constrained Decoding." *Submitted to the International Conference on Learning Representations (ICLR)*, 2027.

## Technical Skills

**Computer Vision:** 3D reconstruction, neural rendering, novel-view synthesis, visual SLAM.

**ML Frameworks:** PyTorch, TensorFlow, scikit-learn.

**Programming:** Python, JavaScript/TypeScript; familiar with C++, Go.

**Software & Systems:** Docker, AWS, SQL, FastAPI, NestJS.

## Academic Service

**Teaching Assistant:** Data Mining (COMP4040).

**Reviewer:** ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM).

**Invited Talk:** *3D Gaussian Splatting*, CAIR Monthly Reading Group, VinUniversity (Mar 19, 2026).

[†]Equal contribution.

**Conference Volunteer:** HORIZONS 2026: Optimization for Sustainability, VinUniversity, Hanoi.

## Industry Experience

**Software Engineer**, *Fullerton Health Vietnam* — Vietnam, Dec 2023 – Oct 2025

- Built healthcare backend services using a microservices architecture.
- Implemented single sign-on and role-based access control; optimized database queries.

**Frontend Engineer**, *Zigvy Corporation* — Vietnam, Apr 2023 – Oct 2023

- Built React UI components and implemented client-facing web application features.

## Achievements

**HCM AI Challenge 2024** — Finalist; developed a web-based event-retrieval pipeline.

**AI Vietnam Program (Class of 2024)** — Completed training in AI and data science.

**Deep Learning Specialization** — DeepLearning.AI (Coursera)

**Applied Data Science Lab** — WorldQuant University

**AWS Certified Cloud Practitioner** — Amazon Web Services